



Fig. 9: UI output



Fig. 10: Various glove movements

Arduino UNO. If this is the first time the Arduino is being connected to the computer, Arduino drivers must be installed first.

7. Deploy LIFA firmware to Arduino. LIFA uses open source Arduino firmware to communicate with LabVIEW. The firmware source code is included within LIFA installer. Once the firmware is deployed to Arduino, it will remain until Arduino is reprogrammed via Arduino IDE.

8. Run LIFA Blink example. Blink example is a simple LabVIEW application that enables the LED attached to Arduino pin 13 to blink. It will confirm whether the hardware and software are set up correctly.

Working of the sixth-sense player

The sixth-sense player uses both LabVIEW and Arduino software. Programs are combined to a single set-up (.exe) file so that you can install this player application on any computer even without LabVIEW and Arduino software. In addition to these two software, an accelerometer is used as the motion sensor. The sensor is used to get different electrical signals as per the movement and position of the sensor attached to the hand glove.

Either you enclose Arduino and the sensor in a suitable enclosure or you may use the hand glove to control the sixth-sense player. We have

EFY Note

The source code, set-up file and testing procedure of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

used an ordinary leather hand glove. Leather is carefully cut so that Arduino and the sensor can be inserted between the leather and linen.

Now, close the opening using adhesive tape or stitch it, so that Arduino and the sensor will not come out. Make another cut on the leather so that you can connect the USB cable to Arduino. The glove used during testing is shown in Fig. 6.

Put on the glove on your hand and connect the USB cable to your PC, and you are ready to play the sixth-sense player.

This project is developed using graphical programming in LabVIEW. You can open the LabVIEW program (sixth sense player.vi) after installing all required packages mentioned in the interfacing steps. Front Panel of this LabVIEW project is shown in Fig. 7 and part of its Block Diagram is shown in Fig. 8.

A set-up program is included in this month's EFY DVD. Run this program to install sixth sense player.exe in your system. Run sixth-sense player.exe program and you will get UI output as shown in Fig. 9. First load the media (video/music) file by clicking on the folder icon and then select the appropriate com port. Different hand glove movements to control the different functions of the media player are shown in Fig. 10.

Future application

It is possible to develop wireless control for this application, so the user can access the application from anywhere in the room. ●



Antony Edison is a student of BE (mechatronics) at Hindustan College of Engineering and Technology. He and his team have successfully designed an electric solar vehicle